FEDERAL SPECIFICATION

INTRAVENOUS SOLUTIONS; 1000 MILLILITER CONTAINER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers intravenous solutions in two types and two classes of specially designed 1000 milliliter containers.

1.2 Classification.—Containers shall be of the following types and classes, as specified (see 6.3).

Type I.—Vacuum type.

Class 1.—With diaphragm stopper.

Class 2.—With perforated stopper.

Type II.—Nonvacuum type.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

PPP–B–585—Boxes; Wood, Wirebound.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN–P–108—Boxes, Fiberboard, Corrugated and Solid (Military Type).

MIL–I–16927 — Intravenous Injection Set.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication forms a part of this specification.

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Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

U. S. Pharmacopoeial Convention Publication:

Pharmacopoeia of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.).

3. REQUIREMENTS

3.1 Intravenous solutions. — Intravenous solutions shall be as specified and shall be sterile, pyrogen-free and conform to applicable requirements for injections of the current edition of the Pharmacopoeia of the United States. Solution shall be clear and substantially free from turbidity, or undissolved or particulate matter, when examined, without accessory magnification (except for such optical correction as may be required to establish normal vision), against a black background and against a white background with illumination from a light which, at a point 25.4 centimeters (10 inches) from its source, provides an intensity of illumination of not less than 100 and not more than 350 footcandles.

3.1.1 *Certification of sterility.*—When specified, a certificate of sterility, listing all lot or control numbers, stating that samples from each sterilizer charge have been tested and found to be sterile, shall be submitted by the contractor to the procuring activity.

3.1.2 *Pyrogens.*—When specified, contractor shall submit a certificate of freedom from pyrogens covering each lot, with lot or control numbers indicated.

3.2 Bottle.

3.2.1 *Material.*—The bottle shall be of type I or IV clear glass as specified in the current edition of the Pharmacopoeia of the United States.

3.2.2 *Design.*—The bottle shall be a cylindrical flask of uncolored clear glass, sufficiently heavy, durable, and of suitable design for the purpose intended. Bottle shall have an overflow capacity of approximately 1200 cubic centimeters. Graduations shall be molded, etched, or baked on the bottle in such a manner that the total volume in the righted position and the dispensed volume in the inverted position can be determined at a glance. Graduations shall be shown in cubic centimeters and they shall be placed at 100 cubic centimeter intervals, except that such graduation marks as may be interfered with by the suspension bail band or shoulder may be omitted.

3.2.2.1 *Suspension bail.*—The bottom of each bottle shall be equipped with a free-swing metal suspension bail fastened securely to the flask by a metal band, or by other suitable means to allow the bottle to be suspended freely but securely in an inverted position. The bail shall be so attached as to tend to prevent inadvertent removal or detachment during use or shipment.

3.2.2.2 *Closure.*

3.2.2.2.1 *Closure for type I, class 1 container.*—The closure for type I, class 1 container shall consist of a 1-piece rubber stopper formed in such a manner that it shall maintain the specified amount of vacuum without the application of the metal seal. The rubber stopper shall be made of fresh stock, high grade, nonblooming, nonbleeding, nonoxidizing rubber which has been properly processed for the purpose. The bottom surface of the stopper shall be coated to reduce contact of the solution with the rubber. The surfaces of depressions in the stopper are not required to be coated. The coating shall be nontoxic, smooth, uniform, flexible, and closely adherent to the stopper, and shall not flake off into the solution in storage nor shall it flake off into solution on needle puncture. The coating shall show no material change from its original properties, and shall not peel or crack off when laterally compressed, nor shed, nor show material blistering after the coated stopper has been autoclaved in distilled water

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at 15 pounds steam pressure for 8 hours. The stopper shall be so designed that it shall have a plug insert which forms a tight fit into the neck of the bottle for at least one centimeter. The rubber stopper shall be so constructed as to provide three integrally molded rubber diaphragms in the upper portion of the stopper. Two of the diaphragms shall be not less than 0.9 millimeters and not more than 1.7 millimeters thick, the location of which shall be clearly designated on the outer surface of the stopper by identifying marks. One of these sealing diaphragms shall be suitable for the accommodation of the airway tube specified in 3.2.2.3. The other diaphragms shall be so designed as to accommodate a standard connector or the connecting device on an injection set. The third integrally molded diaphragm shall be not less than 2.5 and not more than 9.0 millimeters thick, the location of which shall be clearly designated by an identifying mark. Stopper shall pass the leakage test as specified in 4.3.2. Additional markings for identification purposes may be used on the stopper, provided that the additional markings are not such as to confuse the user.

3.2.2.2.2 *Closure for type I, class 2 container.*—The closure for type I, class 2 container shall consist of a rubber stopper with auxiliary or integral devices so as to maintain the specified amount of vacuum. The rubber stopper shall be made of fresh stock, high grade, nonblooming, nonbleeding, nonoxidizing rubber which has been properly processed for the purpose. The bottom surface of the stopper may be partially or wholly coated to reduce contact of the solution with the rubber. If used, the coating shall meet the requirements specified in 3.2.2.2.1. The stopper shall be so designed that it shall have a plug insert which forms a tight fit into the neck of the bottle for at least one centimeter. The rubber stopper shall be provided with suitable outlet and airholes, the former so designed as to accommodate a standard connector or the connecting device on an injection set, and the latter suitable for the accommodation of the airway tube specified in 3.2.2.3. A diaphragm, suitable for penetration by a hypodermic needle, may also be provided, but is not required.

3.2.2.2.3 *Closure for type II container.*—The closure for type II container shall provide an airtight, leakproof seal for the container and its contents. It shall consist of an aluminum cap, to be individually threaded onto each container, complete with an interior aluminum disk, a rubber gasket, and a liner facing. The aluminum disk shall be approximately 0.007-inch thick and shall be so designed as to insure easy removal of the cap. The rubber gasket shall be not less than $\frac{1}{16}$-inch thick manufactured from fresh stock, high grade, nonblooming rubber which has been properly processed for the purpose. The liner facing shall be of gilsonite or other suitable material; shall be nontoxic, smooth, and flexible, and shall not tear nor adhere to the lip of the bottle on removal of the cap. The liner facing shall have no appreciable deleterious effect on the solution and shall show no appreciable material change from its original properties after having been autoclaved in distilled water at 15 pounds steam pressure for 8 hours. The closure shall be so designed that on removal of the cap the component parts shall be removed as a unit. Markings for the identification of the solution shall be stamped, embossed, or otherwise permanently provided on the top surface of the cap.

3.2.2.3 *Airway tube for type I container.*—The airway tube for type I container shall be a glass tube with an inside diameter of approximately 3 millimeters extending from immediately under one of the diaphragms or openings of the rubber stopper to within not more than 10 millimeters of the bottom of the bottle to allow displacement of fluid with air as the flask is emptying. The airway tube shall be of type I or type IV clear glass as described in the current edition of the Pharmacopoeia of the United States.

3.2.2.4 *Seal.*

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3.2.2.4.1 *Seal for type I, class 1 container.*—The rubber stopper for type I, class 1 container shall be held firmly in the bottle by a rolled-on metal band which overlaps the top edge of the stopper but does not cover the diaphragms. The diaphragm portion shall be covered with a metal cap or metal disk overlapping the metal band and held in place with a metal tear-off seal. As an alternate for the metal tear-off seal, a tight fitting metal cap which is readily removed, may be used. A one-piece metal closure seal with central tear-out disk is also acceptable, provided that a rubber, plastic, or cellophane disk larger than the tear-out portion of the metal seal is placed between the seal and the rubber stopper. One drop of U.S.P. glycerin containing 0.4 percent propyl parasept and 0.06 percent chlorothymol may be placed on the top of the rubber stopper if a flexible disk is used. With the cap or overlapping metal disk type of seal the use of flexible disks with or without antiseptic solution are optional.

3.2.2.4.2 *Seal for type I, class 2 container.*—The rubber stopper for type I, class 2 container shall be held firmly in the bottle by a rolled-on metal band which overlaps the top edge of the stopper but does not cover the openings in the stopper. The central portion of the stopper shall be covered with a metal cap or metal disk overlapping the metal band and held in place by a metal tear-off seal. As an alternate for the metal tear-off seal, a tight fitting metal cap which is readily removed, may be used. A 1-piece metal closure seal with central tear-out disk is also acceptable.

3.2.2.4.3 *Seal for type II container.*—The closure for type II container shall be protected against accidental removal or loosening by a rolled aluminum overseal which may be readily removed prior to use of the solution.

3.2.3 *Vacuum for type I container.*—Each filled container for type I shall have a vacuum of not less than 15 inches of mercury, mechanically induced in the flask before stoppering and autoclaving.

3.3 Labeling.

3.3.1 *Labeling of type I container.*—Labeling of each type I container shall include the stock number, nomenclature, quantity, manufacturer's name or registered trade-mark, and address, the manufacturer's lot number, date of manufacture and the words "Sterile" and "Non-Pyrogenic", the caution statement required by the Durham-Humphrey Law, and any other information as may be required by State and Federal laws. In addition, the label shall bear the following statements: "Must be substantially free of any turbidity or undissolved material which can be readily detected without accessory magnification" and "Do not use if no vacuum is detectable". The label shall be of sufficient size to make all lettering clearly legible and it shall be affixed to the bottle with waterproof adhesives to pass test specified in 4.3.3. Each bottle shall have, in addition, a legend, permanently molded, stamped, etched, or baked on the glass or on top of the rubber stopper, or on the rubber stopper covering, stating the solution contained in the bottle (for example, 5 percent dextrose in normal saline). In lieu of a legend, an abbreviation easily recognized and interpreted by the professional user may be substituted, when approved by the contracting officer.

3.3.2 *Labeling of type II container.*—Labeling of type II container shall meet the requirements as specified in 3.3.1, except the label shall not bear the statement "Do not use if no vacuum is detectable."

3.3.3 *Special labeling requirements for Water for Injection.*—In addition to the requirements specified in 3.3.1 or 3.3.2, the labeling of bottles of Water for Injection shall be in accordance with the following requirements:

(a) The name "WATER FOR INJECTION, U.S.P." shall appear.

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(b) The name "DISTILLED WATER" shall not be shown on the label.

(c) The word "SOLUTION" shall not appear on the label.

(d) The following statement shall appear in red ink using upper case type:

"NOT FOR INJECTION IN PRESENT FORM; SEE CAUTION STATEMENT BELOW."

(e) The following statement shall appear in red type: "CAUTION.—If contents are to be used for parenteral therapy, add dextrose or salt and use within 3 hours after opening."

Note.—Suitable variations in the format and wording of the label statements required in (d) and (e), above, will be acceptable, provided that prior approval of the proposed revision is obtained from the contracting officer.

3.4 Workmanship.—The container and its contents shall be free from any defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.3). Moreover, in the absence of specific instructions to the contrary, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at any point of delivery, as specified (see 6.3). In case of factory inspection, every facility shall be afforded inspectors by the manufacturer for the prosecution of their work.

4.3 Tests.

4.3.1 *Intravenous solutions.*—Unless otherwise specified, tests on the intravenous solutions shall be conducted in accordance with the applicable requirements of the current edition of the Pharmacopoeia of the United States.

4.3.2 *Leakage test for type I, class 1 container.*—Insert the connector of a standard intravenous injection set into the outlet diaphragm of the stopper of a filled and sterilized bottle to be tested. The stopper shall not leak around the connector when the bottle is inverted and subjected to an internal pressure of 5 pounds per square inch.

4.3.3 *Label adhesion.* — Labeled bottles taken at random shall be allowed to dry at room temperature for a period of not less than 5 days. The labeled bottles shall be completely submerged in tap water maintained at a temperature of 75° ± 5°F. for a period of not less than 48 hours. Bottles shall then be removed from the water. Complete failure of adhesion on the bottles shall constitute a rejection.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1).

5.1 Packaging.—Bottles shall be packaged in standard commercial containers of the size and kind commonly used for the purpose which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.—The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of

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delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Interior packages.*—Each package shall be marked to identify contents.

5.3.2 *Shipping containers.*—Shipping containers shall be marked with stock number, name of material, quantity, name of consignee, name of consignor, and procurement instruction and office identifying numbers as shown on contract or purchase order.

5.4 Preparation for delivery (Military).

5.4.1 *Packaging.*

5.4.1.1 *Unit of issue.*—Six (6) bottles, as specified, constitutes one unit.

5.4.1.2 *Level A.*—Each unit (6 bottles) shall be packaged in a regular slotted single-wall corrugated fiberboard box of not less than 200-pound Mullen test, with "A" or "C" flute corrugations, meeting all requirements of Federal Specification LLL–B–631. The interior of each box shall be fitted with full size top and bottom pads, and three full height scored shells. The shells shall be scored and folded to form two individual cells or one double shell. All corrugations shall be horizontal. Shells shall provide two (2) thicknesses of combined fiberboard between adjacent glass containers and one thickness between glass containers and the inside faces of the ends and sides of the box. Scored sheets and pads shall be fabricated of the same material as the box. Bottles shall fit snugly in cells.

5.4.1.3 *Level C.*—Packaging shall be as specified in 5.1.

5.4.2 *Packing.*

5.4.2.1 *Level A.*

5.4.2.1.1 *Master carton.*—Two unit cartons (2 units) shall be packed in a fiberboard box conforming to Military Specification JAN–P–108, compliance symbol V3c. Closure shall comply with appendix thereto.

5.4.2.1.2 *Exterior container.*—One master carton (2 units) shall be overpacked in an exterior container designed for type 3 load. Construction and closure shall be in accordance with Federal Specification PPP–B–621 or PPP–B–585, style 3, for class 3 use. Resawn lumber conforming to group III or group IV shall be used in fabrication of wirebound boxes.

5.4.2.1.2.1 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification. Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.4.2.2 *Level B.*—Two unit cartons (2 units) shall be packed in a corrugated fiberboard box conforming to table II of Federal Specification LLL–B–631. Carton shall be adequately secured to prevent accidental opening.

5.4.2.3 *Level C.*—Packing shall be as specified in 5.2.

5.4.3 *Marking.*

5.4.3.1 *Bottles.*—Each bottle shall be labeled as specified in 3.3.

5.4.3.2 *Unit container.*—Each unit container shall be marked with stock number, nomenclature, unit, quantity, date of manufacture, caution statement as required by Durham-Humphrey Law, lot or control number designated as such, "Lot_____" or "Control_____", "Stock No._____", office identifying number, and name of contractor.

5.4.3.3 *Shipping container.*—Shipping container shall be marked in accordance with

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Military Standard MIL–STD–129, except that date cured (for rubber components) is not required. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments for the Medical Services of the Armed Forces.

5.4.3.3.1 *Special markings.*—In addition to the above specified markings, each shipping container shall bear the legend:

"DO NOT PERMIT TO FREEZE"

5.4.3.4 *Names of items.*—The nomenclatures used for identification purposes shall be as specified in 6.4.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and application of the various levels of packaging protection for supplies and equipment.

6.2 Intended use.

6.2.1 *Type I containers.*—Type I containers are intended for use in packaging all items listed in 6.4. The rubber stoppers provided on these containers are so designed as to be suitable for use with the standard military intravenous injection sets (see Military Specification MIL–I–16927), for the administration of intravenous solutions.

6.2.2 *Type II containers.*—Type II containers are intended for use in packaging all items listed in 6.4. The containers are so designed as to be suitable for use with the supplying manufacturer's intravenous injection set.

6.3 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and class required (see 1.2).

(*c*) Intravenous solutions required (see 3.1).

(*d*) Whether certificates of sterility and non-pyrogenicity are required (see 3.1.1 and 3.1.2).

(*e*) Sampling required (see 4.1).

(*f*) Inspection required (see 4.2).

(*g*) Place of final inspection (see 4.2).

(*h*) Level of protection required (see 5.4.1, 5.4.2, and 5.4.3).

6.4 This specification covers the following items appearing in the Armed Services Medical Stock List:

6505–116–4600 Dextrose Injection, USP, 5%, 1000 cc., 6s.

6505–116–4610 Dextrose Injection, USP, 10% 1000 cc., 6s.

6505–116–5000 Dextrose and Sodium Chloride Injection, USP, 1000 cc., 6s.

6505–138–4610 Protein Hydrolysate Injection, USP, 1000 cc., 6s.

6505–149–1720 Water for Injection, USP, 1000 cc., 6s.

6505–153–8651 Sodium Chloride Injection, USP, 1000 cc., 6s.

6505–153–8755 Sodium Lactate Injection, USP, 1/6 Molar, 1000 cc., 6s.

6.5 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.6 Transportation description. — Transportation description applicable to this item is:

Solutions, Intravenous, Saline.
Carload minimum weight 36,000 pounds.

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Solutions, intravenous, nutritional or anticoagulant.
Truckload minimum weight 36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents each.

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